LENA DURAISAMY SWAMIKANNAN, Ph.D.

lxd200013@utdallas.edu ◇ ◇ linkedin.com/in/lenavijay-ai-in-healthcare

OBJECTIVE

Healthcare AI researcher focused on developing and translating artificial intelligence solutions for real-world clinical and public health challenges. Expertise in medical computer vision, federated learning, and analytics across experimental and clinical datasets, with experience building end-to-end Medical AI pipelines from problem formulation, data curation, preprocessing, and model development to validation, calibration, deployment, and real-time inference. Research interests center on clinically deployable and privacy-preserving AI systems that address healthcare disparities, improve health equity, support clinical decision-making, enable early cancer detection, and expand access to high-quality care.

RESEARCH SKILLS

Medical AI	Medical Image Analysis, Mobile Vision, Federated Learning
Edge AI Deployment	Resource-Constrained AI, Offline AI Systems
Healthcare AI Product Development	End-to-End Medical AI Pipelines, Clinical Translation, Digital Health

WORK EXPERIENCE

Researcher, Quality of Life Technology Lab, The University of Texas at Dallas Jan 2021 – Aug 2026
Erik Jonsson School of Engineering and Computer Science

- Built a customized clinical oral-image database and designed an end-to-end, real-time federated training workflow for privacy-preserving oral cancer screening.
- Established cross-institutional federated learning workflows with external research sites to comply with data privacy requirements, enabling decentralized model training across Dallas and Philadelphia without sharing raw patient data and supporting development of a more generalizable global Medical AI model.
- Developed mobile-friendly, lightweight AI models for oral cancer detection using pretrained architectures and transfer learning to address limited clinical data availability, enabling efficient deployment in resource-constrained and federated learning environments.
- Evaluated cross-device federated learning performance on NVIDIA Jetson Orin edge platforms and translated the above findings into Android and iOS smartphone applications for fully on-device oral cancer screening, demonstrating the feasibility of an offline, AI-enabled screening tool designed to improve healthcare accessibility in remote and underserved settings.

Graduate Teaching Assistant, The University of Texas at Dallas Jan 2022 - Aug 2026
Erik Johnsson School of Engineering and Computer Science

- Supported instruction for core undergraduate and graduate engineering courses, including Wireless Sensor Networks and Introduction to Digital Circuits, across multiple semesters by assisting both in-person and online course delivery, laboratory coordination, demonstrations, troubleshooting, and preparation of instructional materials.
- Provided continuous academic support through weekly office hours, mentoring, email communication, concept clarification, problem-solving guidance, and individualized assistance, while developing instructional videos and supplementary resources as part of laboratory coursework development.
- Managed extensive assessment and course administration responsibilities, including grading assignments, laboratory reports, exams, and experimental work; maintaining accurate grade records; providing constructive feedback; handling regrade-related issues; and ensuring consistency, fairness, and academic integrity.

UTD Summercamp Student Assistant, The University of Texas at Dallas June 2024 - August 2024
Erik Johnsson School of Engineering and Computer Science

- Developed and delivered a structured, age-appropriate curriculum for high school students as part of the "AI in Healthcare" summer camp, simplifying complex AI concepts through interactive and engaging instruction.

- Conducted hands-on laboratory sessions covering the end-to-end AI development pipeline, including medical image preprocessing, model development, training, evaluation, and deployment into AI-based applications using diverse healthcare datasets.
- Mentored students through individual AI projects, provided technical guidance throughout project development, and designed quizzes and interactive assessments to evaluate understanding and reinforce key learning outcomes.

Research Scholar, Speech Lab, SSN college of Engineering June 2017 - April 2018
Electronics and Communication Engineering

- Developed a noisy Tamil speech corpus by adding five selected non-stationary noises from the NOISEX-92 database to clean Tamil speech across multiple SNR levels, implemented speech enhancement algorithms, and evaluated speech recognition performance on the enhanced speech.
- Investigated speaker spoofing attacks in Automatic Speaker Verification (ASV) systems using the ASVspoof 2015 database, with emphasis on speaker impersonation and vulnerabilities in voice-based biometric authentication.
- Studied anti-spoofing countermeasures for biometric authentication systems and explored deep learning-based approaches to distinguish genuine speech from spoofed speech and improve the security and reliability of speaker verification.

EDUCATION

Ph.D. in Electrical Engineering 2021 - 2026
The University of Texas at Dallas, Richardson, TX. CGPA : 3.66/4

M.E. in Communication Systems 2015 - 2017
Sri Sivasubramaniya Nadar College of Engineering, Chennai. CGPA : 8.70/10

B.E. in Electronics and Communication Engineering 2011 - 2015
Ultra College of Engineering and Technology for Women, Madurai. CGPA : 8.33/10

RESEARCH PUBLICATIONS

1. **Lena D. Swamikannan**, Prajwal Mahesha Sethurani, Akshay Bhagwan Sonawane, C. S. Mani, Jay S. Patel, Prabha Sundaravadivel, and Lakshman Tamil, "Federated Edge Learning on Resource-Constrained Devices: Building an AI Model for Oral Cancer Detection", *in 2nd Smart Medicine and Privacy Protection Workshop, IEEE Bioinformatics and Biomedicine (BIBM)*, Dallas, TX, USA, 2026 (Accepted).

2. **Lena D. Swamikannan**, A. B. Sonawane, J. S. Patel, C. S. Mani, L. Narayana, and L. Tamil, "A Federated Learning Framework for Privacy-Preserving Oral Cancer Screening on Smartphones," *IEEE Journal of Biomedical and Health Informatics* (under review). arXiv:2608.19462.

3. **Lena D. Swamikannan**, A. B. Sonawane, J. S. Patel, C. S. Mani, L. Narayana, and L. Tamil, "Oral Cancer Detection Using Mobile Vision Technology," in *IEEE-EMBS International Conference on Biomedical and Health Informatics*, Houston, TX, USA, 2024, pp. 1–8. doi:10.1109/BHI62660.2024.10913489.

4. Akshay B. Sonawane, **Lena D. Swamikannan**, and L. Tamil, "Robust Classification of Oral Cancer with Limited Training Data," *arXiv preprint*, 2025. arXiv:2510.01547.

5. **D. S. K. Lena** and P. Vijayalakshmi, "Speech Enhancement in Vehicular Environments as a Front End for Robust Speech Recogniser," in *2017 IEEE International Conference on Intelligent Computing and Control Systems (ICICCS)*, Madurai, India, 2017, pp. 430–435. doi:10.1109/ICCONS.2017.8250759.

6. A. Premalatha, **D. S. K. Lena**, S. Lakshmipriya, and R. Lakshmipriya, "Performance Enhancement of VANET Using EMAP," *International Journal of Advanced Information Science and Technology (IJAIST)*, vol. 35, no. 35, pp. 163–166, Mar. 2015. [ResearchGate]

CERTIFICATIONS

Lena D. Swamikannan, Ph.D. lxd200013@utdallas.edu | | LinkedIn

AWS Certified AI Practitioner 2026
AI Infrastructure and Operation Fundamentals- NVIDIA 2026
Graduate Teaching Certificate- Center for Teaching and Learning at The University of Texas at Dallas 2024
Neural Networks and Deep Learning- DeepLearning.AI. 2023
Silver, Gold, Platinum Certified Engineer- BSNL, India 2014

ACADEMIC AWARDS

University Rank holder in Master Of Engineering 2017
Department Rank 2 (M.E.) – SSN College of Engineering, Chennai 2017
Gold Medalist in Anna University Third Semester Examination 2016
Department Rank 1 (B.E.) – Ultra College of Engineering and Technology for Women, Madurai 2015
Appreciation Award – DCAS 2022, Dallas, Texas 2022
NSF Travel Award – IEEE BHI 2024, Houston, Texas 2024

PROFESSIONAL SERVICE

Session Co-Chair, DCAS 2022 2022
Program Committee Reviewer Medical Reasoning with Generative AI Workshop, NeurIPS 2026 2026

LANGUAGE SKILLS

Languages: English, Tamil (fluent in reading, writing, and speaking), Hindi (fluent in reading and writing)